

Distributed Extended Object Tracking Algorithm Based on Contour Measurement Model

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Abstract—We develop a computationally efficient Distributed Extended Object Tracking algorithm based on a contour measurement model. We utilize a diffusion strategy for information fusion in the distributed network. Moreover, each node locally executes an alternating minimization scheme to jointly estimate the kinematic and shape states via Maximum Likelihood Estimation (MLE) and the Iterated Extended Kalman Filter (IEKF). Simulation results verify the superior performance and computational efficiency of the proposed algorithm.

Index Terms—Extended object tracking, Distributed estimation, diffusion strategy, Alternating minimization, Contour measurement model

I. INTRODUCTION

With the widespread application of high-resolution sensors, a single target generates multiple measurements at each time step, which inherently contain extended information such as shape and orientation [1]. Traditional point target tracking methods disregard spatial dimensions and thus are fundamentally incapable of estimating the extended states of an object. In contrast, Extended Object Tracking (EOT) exploits these measurements to jointly estimate kinematic and extended states. This capability enables EOT to effectively handle complex scenarios and has led to its widespread application in maritime surveillance and vehicle tracking [2]–[4].

Within the EOT framework, appropriate modeling of both the object shape and the spatial distribution of measurements is fundamental to achieving effective tracking [3], [5]. Common shape representations include Random Hypersurface Models (RHM) and simple geometric shapes such as ellipses and rectangles [1], [6]. Among these, the elliptical model is widely adopted due to its mathematical tractability and effectiveness in approximating vehicle shapes, and is therefore the focus of this work. Regarding the spatial distribution of measurements, models are generally categorized into surface model and contour model, where scattering centers are assumed to be distributed over the entire object surface and along the object

contour, respectively [2], [3], [5]. Under the surface model assumption, Random Matrix (RM) [1], [7] approaches and the Multiplicative Error Model (MEM) [4], [8] are widely applied for extended state estimation and modeling. For contour models, the Gaussian Mixture Model (GMM) [3], [6], [8] is a prevalent approach. To enhance its estimation performance, the Virtual Measurement Model (VMM) [2] generates artificial data to enforce geometric constraints, yet this approach introduces a significant computational burden, limiting its real-time applicability.

Distributed Extended Object Tracking (DEOT) has attracted significant attention due to its superior tracking performance [9]–[11]. Representative algorithms include the Distributed Random Matrix (DRM) [9], which leverages weighted Kullback-Leibler averaging, and the DEOT Kalman filter (DEOKF) [10], which employs the adapt-then-combine (ATC) diffusion strategy for heterogeneous networks. Additionally, [11] developed a Measurement Combination (MC) estimator to handle multisensor fusion. Despite distinct fusion mechanisms, these works primarily rely on surface model assumptions. Specifically, we select DEOKF [10] for comparison, as it employs the same ATC diffusion strategy as our proposed method. This choice ensures that performance differences can be directly attributed to the measurement model, rather than differences in the distributed fusion strategy.

In this work, we propose a computationally efficient Distributed Contour-based Extended Object Tracking (DCEOT) algorithm. This method explicitly addresses the estimation inaccuracy of traditional surface-based DEOT approaches when measurements are distributed along the object contour. We enhance tracking performance through an ATC diffusion strategy, while simultaneously ensuring low computational complexity via an alternating minimization scheme. Simulation results verify the superior performance and computational efficiency of the proposed algorithm.

II. PROBLEM FORMULATION

We herein consider tracking an extended object over a distributed network $\mathcal{V}$ of N nodes. Two nodes are defined

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as neighbors if they can communicate directly via a communication link [10], [12], [13]. We denote the neighborhood of each node k , which includes each node k itself and all its neighbors, as $\mathcal{V}_k$. The cardinality of this set is defined as the degree of each node k , denoted by d_k , as shown in Fig. 1.

A. Dynamic Model

We consider approximating the shape of the extended object with an ellipse [2], [6], [14], [15]. The extended state vector at each time instant t , incorporating the kinematic and shape states, is defined as

$$\mathbf{u}_t \triangleq [\mathbf{x}_t^\top \dot{\mathbf{x}}_t^\top l_{1,t} l_{2,t}]^\top, \quad (1)$$

where $\mathbf{x}_t = [x_{1,t}, x_{2,t}]^\top$ denotes the position in $\mathbb{R}^2$, $\dot{\mathbf{x}}_t = [\dot{x}_{1,t}, \dot{x}_{2,t}]^\top$ represents the velocity, and $l_{1,t}$ and $l_{2,t}$ correspond to the semi-major and semi-minor axes of the elliptical object, respectively, as shown in Fig. 2.

The state translation equation is given by [2], [6], [14]–[16]

$$\mathbf{u}_{t+1} = \mathbf{F}_t \mathbf{u}_t + \mathbf{w}_t, \quad (2)$$

where $\mathbf{F}_t$ is the known state transition matrix, and $\mathbf{w}_t$ is a zero-mean white Gaussian process noise with covariance matrix $\mathbf{Q}_t$.

B. Measurement Model

Consider the contour-based measurement model for an elliptical extended object. We assume that the measurements are generated from scattering sources distributed along the contour of the elliptical object [2], [14], [17], [18]. As illustrated in Fig. 1, each measurement corresponds to a point on the object perimeter, perturbed by additive white Gaussian noise. The measurement of the scattering source j from each node k at time t is modeled as [6]

$$\mathbf{y}_{k,t}^{(j)} = \mathbf{H} \mathbf{u}_t + \text{Rot}(\alpha_t) \mathbf{s}_t(\theta_j) + \mathbf{v}_{k,t}^{(j)}, \quad (3)$$

where $\mathbf{H} = [\mathbf{I}_2, \mathbf{0}]$ is the measurement matrix, $\mathbf{H} \mathbf{u}_t$ denotes the position of the object center in the global Cartesian frame, and $\mathbf{v}_{k,t}^{(j)}$ is the additive white Gaussian noise with zero mean and covariance $\mathbf{R}_k$. The rotation matrix $\text{Rot}(\alpha_t)$ implements a rotation based on the object orientation $\alpha_t = \text{atan2}(\dot{x}_{1,t}, \dot{x}_{2,t})$ [6], [16]. The vector $\mathbf{s}_t(\theta_j)$ represents the position of the scattering source within the local canonical frame of the target, given by

$$\mathbf{s}_t(\theta_j) = [l_{1,t} \cos(\theta_j), l_{2,t} \sin(\theta_j)]^\top \quad (4)$$

where θ_j denotes the angular parameter of the scattering source j , which is assumed to be uniformly distributed over the interval $[0, 2\pi)$ [6], [17].

We herein define the collection of local measurements at each node. At each time instant t , each node k obtains all $a_{k,t}$ measurements from scattering sources, $\mathcal{Y}_{k,t} \triangleq \{\mathbf{y}_{k,t}^{(1)}, \mathbf{y}_{k,t}^{(2)}, \dots, \mathbf{y}_{k,t}^{(a_{k,t})}\}$. Furthermore, let $\mathcal{Y}_{k,1:t} \triangleq \{\mathcal{Y}_{k,1}, \mathcal{Y}_{k,2}, \dots, \mathcal{Y}_{k,t}\}$ denote all local measurements up to time t at each node k .

In the distributed network, it is assumed that the measurements $\mathcal{Y}_{k,t}$ and $\mathcal{Y}_{l,t}$ obtained at any two distinct nodes k and

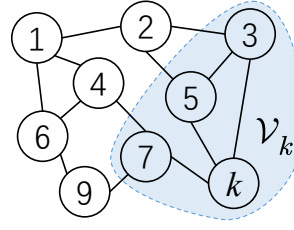


Fig. 1: Topology

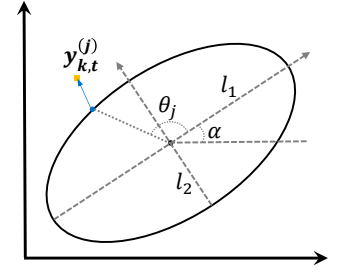


Fig. 2: Measurement model

l ($k \neq l$) are uncorrelated [10]. Each node k has access to the measurements within its neighborhood $\mathcal{V}_k$. We aggregate these neighbors measurements

$$\mathcal{Z}_{k,t} \triangleq \{\mathcal{Y}_{r_1,t}, \dots, \mathcal{Y}_{r_{d_k},t}\} = \{\mathbf{z}_{k,t}^{(1)}, \mathbf{z}_{k,t}^{(2)}, \dots, \mathbf{z}_{k,t}^{(b_{k,t})}\}, \quad (5)$$

where $r_1, r_2, \dots, r_{d_k}$ are the indices of all neighbors within $\mathcal{V}_k$, and $\mathbf{z}_{k,t}^{(i)}$ denotes the i th column of $\mathcal{Z}_{k,t}$, with $b_{k,t} \triangleq \sum_{r \in \mathcal{N}_k} a_{r,t}$. Furthermore, let $\mathcal{Z}_{k,1:t} = \{\mathcal{Z}_{k,1}, \mathcal{Z}_{k,2}, \dots, \mathcal{Z}_{k,t}\}$ denote all measurements up to time t at each node k .

To capture partial measurements within time instant t , define $\mathcal{Z}_{k,1:t}^j$ as the cumulative measurements up to the measurement j at time instant t

$$\mathcal{Z}_{k,1:t}^j = \{\mathcal{Z}_{k,1:t-1}, \mathbf{z}_{k,t}^{(1)}, \mathbf{z}_{k,t}^{(2)}, \dots, \mathbf{z}_{k,t}^{(j)}\}. \quad (6)$$

The aggregated measurement $\mathcal{Z}_{k,1:t}^j$ forms the local information for the subsequent local state estimation at each node k .

III. ALGORITHM DERIVATION

We develop a distributed extended object tracking algorithm based on contour model (DCEOT) to track elliptical object. The proposed algorithm iterates through three stages at each node. First, localized adaptation computes intermediate estimates from aggregated measurements. Second, the combination fuses these estimates using the ATC strategy. Finally, the time-update predicts the extended state for the next instant.

A. Localized Adaptation

Herein, each node k sequentially processes the collected measurements to update the extended state. Given that the angular parameter θ_j is independent of the extended state $\mathbf{u}_{k,t}$, the joint posterior density of $\mathbf{u}_t$ and θ_j conditioned on the measurement set can be formulated via Bayes' rule as

$$p(\mathbf{u}_t, \theta_j | \mathcal{Z}_{k,1:t}^j) \propto p(\mathbf{z}_{k,t}^{(j)} | \mathbf{u}_t, \theta_j) p(\mathbf{u}_t | \mathcal{Z}_{k,1:t}^{j-1}), \quad (7)$$

where we assume the prior probability follows a Gaussian distribution [19] $p(\mathbf{u}_t | \mathcal{Z}_{k,t}^{j-1}) = \mathcal{N}(\mathbf{u}_t; \hat{\mathbf{u}}_{k,t}^{j-1}, \hat{\mathbf{P}}_{k,t}^{j-1})$. Here, $\hat{\mathbf{u}}_{k,t}^j$ and $\hat{\mathbf{P}}_{k,t}^j$ denote the intermediate posterior state estimate and the corresponding error covariance, respectively, obtained after processing the measurement j at each node k . The sequential update is initialized with the predicted state and covariance from the time update stage, i.e., $\hat{\mathbf{u}}_{k,t}^0 = \hat{\mathbf{u}}_{k,t|t-1}$ and $\hat{\mathbf{P}}_{k,t}^0 = \hat{\mathbf{P}}_{k,t|t-1}$. Let node $l \in \mathcal{V}_k$ denote the source

Algorithm 1: DCEOT

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1: Step 1: Measurement Update
2: Initialize:  $\hat{\mathbf{u}}_{k,t}^0 = \hat{\mathbf{u}}_{k,t|t-1}$ ,  $\hat{\mathbf{P}}_{k,t}^0 = \hat{\mathbf{P}}_{k,t|t-1}$ 
3: for  $j = 1$  to  $b_{k,t}$  do
4:   Initialize iteration:  $\mathbf{u}^{(0)} = \hat{\mathbf{u}}_{k,t}^{j-1}$ 
5:   for  $n = 1$  to  $N_{\max}$  do
6:     for  $l = 1$  to  $M_{\max}$  do
7:       Calculate angular parameter  $\theta_j^{(l)}$  by (10)-(12).
8:     end for
9:     Set angular parameter  $\hat{\theta}_j = \theta_j^{(l)}$ .
10:    Calculate state estimate  $\mathbf{u}^{(n)}$  by (13)-(16).
11:  end for
12:  Set state estimate and error covariance
     $\hat{\mathbf{u}}_{k,t}^j = \mathbf{u}^{(n)}$ ,  $\hat{\mathbf{P}}_{k,t}^j = (\mathbf{I} - \mathbf{K}^{(n)}\mathbf{H}^{(n)})\hat{\mathbf{P}}_{k,t}^{j-1}$ 
13: end for
14: Set local estimates  $\hat{\mathbf{u}}_{k,t} = \hat{\mathbf{u}}_{k,t}^{b_{k,t}}$ ,  $\hat{\mathbf{P}}_{k,t} = \hat{\mathbf{P}}_{k,t}^{b_{k,t}}$ 
15: Step 2: Combination
16: Update local estimates  $\hat{\mathbf{u}}_{k,t}$  and  $\hat{\mathbf{P}}_{k,t}$  via (17)
17: Step 3: Time Update
18: Update predictions  $\hat{\mathbf{u}}_{k,t+1|t}$  and  $\hat{\mathbf{P}}_{k,t+1|t}$  via (18)

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of measurement $\mathbf{z}_{k,t}^{(j)}$. The joint MAP estimation problem is equivalent to minimizing the following cost function

$$\|\mathbf{z}_{k,t}^{(j)} - h(\mathbf{u}_t, \theta_j)\|_{\mathbf{R}_t^{-1}}^2 + \|\mathbf{u}_t - \hat{\mathbf{u}}_{k,t}^{j-1}\|_{(\hat{\mathbf{P}}_{k,t}^{j-1})^{-1}}^2. \quad (8)$$

Here, the notation $\|\mathbf{a}\|_{\mathbf{W}}^2 \triangleq \mathbf{a}^\top \mathbf{W} \mathbf{a}$ and $h(\mathbf{u}, \theta_j) \triangleq \mathbf{H} \mathbf{u} + \text{Rot}(\alpha) \mathbf{s}(\theta_j)$.

We use an alternating minimization strategy [20], [21] to solve this problem. This strategy decomposes the problem into two tractable subproblems, which are solved in turn in each iteration.

1) *Angular Parameter Update:* We herein fix the extended state estimate $\hat{\mathbf{u}}_{k,t}^{j-1}$ and estimate the angular parameter for the incoming measurement. Under the uniform prior assumption, the MAP estimation of θ_j reduces to MLE, and θ_j is obtained by minimizing

$$\mathcal{J}(\theta_j) \triangleq \|\mathbf{e}(\theta_j)\|_{(\mathbf{R}_t)^{-1}}^2, \quad (9)$$

where the error term is defined as $\mathbf{e}(\theta_j) \triangleq \mathbf{z}_{k,t}^{(j)} - h(\hat{\mathbf{u}}_{k,t}^{j-1}, \theta_j)$.

We employ the Newton-Raphson method to solve this problem. The parameter is iteratively updated by

$$\theta_j^{(m+1)} = \theta_j^{(m)} - \mathcal{J}'(\theta_j^{(m)}) / \mathcal{J}''(\theta_j^{(m)}). \quad (10)$$

where the first-order derivative $\mathcal{J}'$ and the second-order derivative $\mathcal{J}''$ are explicitly derived as

$$\mathcal{J}'(\theta) = -\mathbf{e}(\theta)^\top (\mathbf{R}_{k,t}^{(j)})^{-1} \mathbf{r}'(\theta), \quad (11)$$

$$\mathcal{J}''(\theta) = \mathbf{r}'(\theta)^\top (\mathbf{R}_{k,t}^{(j)})^{-1} \mathbf{r}'(\theta) - \mathbf{e}(\theta)^\top (\mathbf{R}_{k,t}^{(j)})^{-1} \mathbf{r}''(\theta), \quad (12)$$

where $\mathbf{r}(\theta) = \text{Rot}(\alpha_t) \text{diag}(l_{1,t}, l_{2,t}) [\cos \theta, \sin \theta]^\top$.

To ensure robust and rapid convergence, the iteration is initialized with $\theta^{(0)} = \text{atan2}(\hat{y}/\hat{l}_{2,t|t-1}, \hat{x}/\hat{l}_{1,t|t-1})$, where $[\hat{x}, \hat{y}]^\top = \text{Rot}(\alpha_{t|t-1})^\top (\mathbf{z}_{k,t}^{(j)} - \mathbf{H} \hat{\mathbf{u}}_{k,t|t-1})$ is derived from

the predicted extended state $\hat{\mathbf{u}}_{k,t|t-1}$, and $\hat{l}_{1,t|t-1}, \hat{l}_{2,t|t-1}$ are the semi-axis components of $\hat{\mathbf{u}}_{k,t|t-1}$. Empirical results confirm consistent convergence within a few iterations. The final estimate upon convergence is denoted as $\hat{\theta}_j$.

2) *Extended State Update:* We herein substitute the estimated angular parameter $\hat{\theta}_j$ into (8) and estimate the extended state $\mathbf{u}_t$. We employ an iterated extended Kalman filter (IEKF) to solve this MAP problem [19], [22]. For notational brevity, we omit the indices k, t in the inner iterations. Following [6], we omit the derivatives of $\text{Rot}(\alpha_t)$ with respect to the extended state $\mathbf{u}$ to avoid severe linearization errors and enhance filter stability. Let n denote the IEKF iteration index, initialized with $\mathbf{u}^{(0)} = \hat{\mathbf{u}}_{k,t}^{j-1}$. The iterative update proceeds as follow

$$\mathbf{H}^{(n)} \triangleq \begin{bmatrix} 1 & 0 & 0 & 0 & \cos \alpha_t \cos \hat{\theta}_j & -\sin \alpha_t \sin \hat{\theta}_j \\ 0 & 1 & 0 & 0 & \sin \alpha_t \cos \hat{\theta}_j & \cos \alpha_t \sin \hat{\theta}_j \end{bmatrix} \quad (13)$$

$$\mathbf{S}^{(n)} = \mathbf{H}^{(n)} \hat{\mathbf{P}}_{k,t}^{j-1} (\mathbf{H}^{(n)})^\top + \mathbf{R}_{k,t}^{(j)}, \quad (14)$$

$$\mathbf{K}^{(n)} = \hat{\mathbf{P}}_{k,t}^{j-1} (\mathbf{H}^{(n)})^\top (\mathbf{S}^{(n)})^{-1}, \quad (15)$$

$$\begin{aligned} \mathbf{u}^{(n)} &= \hat{\mathbf{u}}_{k,t}^{j-1} + \mathbf{K}^{(n)} \left[\mathbf{z}_{k,t}^{(j)} - h(\mathbf{u}^{(n-1)}; \hat{\theta}_j) \right. \\ &\quad \left. - \mathbf{H}^{(n)} (\hat{\mathbf{u}}_{k,t}^{j-1} - \mathbf{u}^{(n-1)}) \right]. \end{aligned} \quad (16)$$

The above iterative process is repeated until convergence. Upon convergence, the final optimized state estimate and error covariance are denoted by $\hat{\mathbf{u}}_{k,t}^j$ and $\hat{\mathbf{P}}_{k,t}^j$, respectively.

It can be observed that the IEKF inner iteration inherently depends on the angular parameter θ via both the Jacobian matrix and the innovation residual. Similar to the strategy employed in [22], we integrate the update of θ directly into the IEKF inner loop, as explicitly detailed in Lines 5-11 of Algorithm 1. This coupled update ensures that the measurement model is linearized at the most probable scattering source, thereby reducing linearization errors induced by θ . The complete measurement update procedure is summarized in the *Measurement Update* part of Algorithm 1.

B. Combination

We now consider the combination step at each node k to enhance the estimation performance. Specifically, adopting the ATC diffusion strategy [23], each node k updates its local estimates by linearly fusing the intermediate results from its closed neighborhood $\mathcal{V}_k$:

$$\hat{\mathbf{u}}_{k,t} = \sum_{l \in \mathcal{V}_k} c_{lk} \hat{\mathbf{u}}_{l,t}^{b_{l,t}}, \quad \hat{\mathbf{P}}_{k,t} = \sum_{l \in \mathcal{V}_k} c_{lk} \hat{\mathbf{P}}_{l,t}^{b_{l,t}}, \quad (17)$$

where the non-negative combination weights c_{lk} satisfy $\sum_{l \in \mathcal{V}_k} c_{lk} = 1$ [23]–[25], and are chosen uniformly for computational simplicity or non-uniformly to optimize estimation.

C. Time-update

Based on the linear state evolution model (2), the time update step propagates the posterior estimates from the current time instant t to obtain the predicted state and error covariance for the next time instant $t+1$

$$\hat{\mathbf{u}}_{k,t+1|t} = \mathbf{F}_t \hat{\mathbf{u}}_{k,t}, \quad \hat{\mathbf{P}}_{k,t+1|t} = \mathbf{F}_t \hat{\mathbf{P}}_{k,t} \mathbf{F}_t^\top + \mathbf{Q}_t. \quad (18)$$

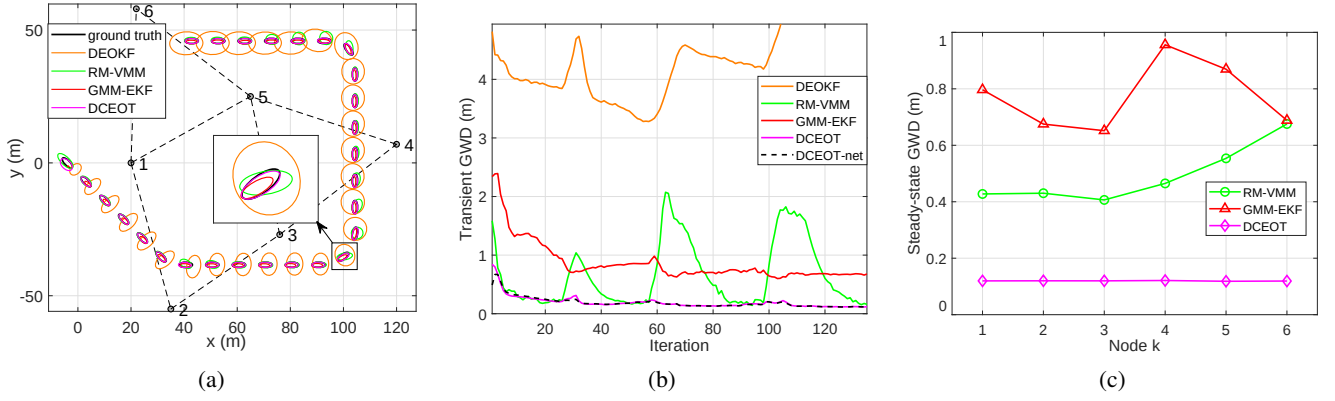


Fig. 3: Performance comparison of different algorithms. (a) Trajectory and estimation at Node 2. (b) Transient GWD at Node 2 and the network average. (c) Steady-state GWD.

The implementation of the proposed algorithm is summarized in Algorithm 1.

IV. SIMULATION RESULTS

We evaluate the proposed DCEOT algorithm against the distributed DEOKF [10], and two methods, GMM-EKF [6] and RM-VMM [2], which operate without neighbor information.

We consider an elliptical extended object with semi-axis lengths of 5 m and 2 m [2], which moves at a constant speed of 36 km/h within a distributed network of 10 sensor nodes. For simplicity, we use uniform combination weights $c_{lk} = 1/d_k$ at each node k . The trajectory comprises one 45° turn and two 90° turns [10], and the network topology as well as the ground truth trajectory are illustrated in Fig. 3a. The measurements are generated following the mechanism described in [2], with the sensor resolution set to 0.2° and the measurement noise covariance matrix given by $\mathbf{R}_k = \text{diag}(0.5^2, 0.5^2)$. In addition, the state transition and process noise covariance matrices are given by

$$\mathbf{F}_t = \begin{bmatrix} \mathbf{I}_2 & \Delta t \mathbf{I}_2 & \mathbf{0}_2 \\ \mathbf{0}_{4 \times 2} & \mathbf{I}_4 \end{bmatrix} \quad (19)$$

$$\mathbf{Q}_t = \text{diag}\left(\frac{\Delta t^3}{6}, \frac{\Delta t^3}{6}, \frac{\Delta t}{2}, \frac{\Delta t}{2}, 0, 0\right), \quad (20)$$

with the sampling interval $\Delta t = 0.2$ s. The initial estimates for the extended state and error covariance matrix at each node k are set to $\hat{\mathbf{u}}_{k,0} = [-15, -5, 5, -5, 2, 1]^\top$ and $\hat{\mathbf{P}}_{k,0} = \text{diag}(5^2, 5^2, 2^2, 2^2, 1, 1)$. The maximum iteration counts $N_{\max}$ and $M_{\max}$ are set to 5 and 3, respectively.

We utilize the Gaussian Wasserstein Distance (GWD) to evaluate the extended object tracking performance [1], [10]. The GWD of each node k at time step t is defined as

$$\text{GWD}_{k,t}^2 \triangleq \|\mathbf{x}_t - \hat{\mathbf{x}}_{k,t}\|^2 + \text{tr}(\mathbf{S}_t + \hat{\mathbf{S}}_{k,t} - 2\sqrt{\sqrt{\mathbf{S}_t}\hat{\mathbf{S}}_{k,t}\sqrt{\mathbf{S}_t}}), \quad (21)$$

where $\mathbf{S}_t$ represents the extent matrix given by $\mathbf{S}_t = \text{Rot}(\alpha_t)\text{diag}(l_{1,t}^2, l_{2,t}^2)\text{Rot}(\alpha_t)^\top$. Additionally, the transient network GWD (denoted as DCEOT-net in Fig. 3b) at time

t as $\text{GWD}_t^{\text{net}} \triangleq \frac{1}{N} \sum_{k=1}^N \text{GWD}_{k,t}$, with steady-state GWDs obtained by averaging the corresponding transient results over 20 iterations after convergence.

We herein evaluate the performance of different algorithms. The estimated trajectories of the elliptical object and the overall tracking performance are illustrated in Fig. 3. Overall, the proposed DCEOT algorithm achieves superior tracking performance. This is attributed to its effective diffusion strategy and the accurate contour measurement model. In contrast, the GMM-EKF exhibits limited accuracy during constant-velocity motion, while the RM-VMM suffers significant degradation during turns. The conventional DEOKF yields poor estimates due to its mismatch with the actual contour-based measurement distribution.

In terms of computational complexity, as shown in Table I, the proposed DCEOT algorithm requires only a slight increase in computation compared to the GMM-EKF, while delivering significantly better tracking accuracy. Moreover, it imposes a substantially lower computational burden than the RM-VMM, which incurs a heavy cost due to the generation of massive virtual measurements.

TABLE I: Ratio of calculation time to the baseline GMM-EKF [6] algorithm

Algorithm	RM-VMM [2]	DCEOT
Relative Runtime	36.38	1.23

V. CONCLUSIONS

This paper addresses the performance degradation of traditional surface-based DEOT methods when measurements follow a contour distribution. We propose a Distributed Contour-based Extended Object Tracking (DCEOT) algorithm to resolve this issue. The algorithm enhances tracking accuracy through ATC diffusion-based information fusion and ensures low computational complexity via an alternating minimization scheme that efficiently couples MLE and IEKF. Simulations confirm that DCEOT achieves superior estimation accuracy and high computational efficiency.

REFERENCES

- [1] K. Granström and M. Baum, "A tutorial on multiple extended object tracking," *Authorea Preprints*, 2023, doi: <https://doi.org/10.36227/techrxiv.19115858.v1>.
- [2] P. Hoher, S. Wirtensohn, T. Baur, J. Reuter, F. Govaers, and W. Koch, "Extended target tracking with a lidar sensor using random matrices and a virtual measurement model," *IEEE Trans. Signal Process.*, vol. 70, pp. 228–239, 2022.
- [3] Z. Wen, J. Lan, L. Zheng, and T. Zeng, "Vehicle tracking using shape-dependent mixture model with edge-concentrated measurements," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 6, pp. 8337–8352, 2025.
- [4] S. Yang and M. Baum, "Tracking the orientation and axes lengths of an elliptical extended object," *IEEE Trans. Signal Process.*, vol. 67, no. 18, pp. 4720–4729, Sep. 2019.
- [5] Z. Li, C. Liu, S. Ma, and H. Wang, "Extended object tracking under a state-coupled model with gaussian mixture distribution," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, 2024, pp. 2890–2897.
- [6] H. Alqaderi, F. Govaers, and R. Schulz, "Spacial elliptical model for extended target tracking using laser measurements, Bonn, Germany," in *2019 Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, 2019, pp. 1–6.
- [7] J. Lan and X. R. Li, "Tracking of extended object or target group using random matrix: new model and approach," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 6, pp. 2973–2989, Dec. 2016.
- [8] K. Thormann, S. Yang, and M. Baum, "Kalman filter based extended object tracking with a gaussian mixture spatial distribution model," in *2021 IEEE Intelligent Vehicles Symposium Workshops (IV Workshops)*, 2021, pp. 293–298.
- [9] W. Li, Y. Jia, D. Meng, and J. Du, "Distributed tracking of extended targets using random matrices, Osaka, Japan," in *Proc. IEEE Conf. Decis. Control*, 2015, pp. 3044–3049.
- [10] Y. Ren and W. Xia, "Distributed extended object tracking based on diffusion strategy, Amsterdam, Netherlands," in *Proc. 28th European Signal Process. Conf.*, 2021, pp. 2338–2342.
- [11] X. Zhang and J. Lan, "Measurement combination estimator for multisensor extended object tracking using random matrix," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 1, pp. 698–715, 2024.
- [12] A. H. Sayed, "Adaptive networks," *Proc. of the IEEE*, vol. 102, no. 4, pp. 460–497, Apr. 2014.
- [13] W. Xia and Y. Zhang, "Resilient distributed estimation against FDI attacks: A correntropy-based approach," *Information Sciences*, vol. 635, pp. 236–256, Jul. 2023.
- [14] P. Hoher, J. Reuter, D. Dold, D. Griesser, F. Govaers, and W. Koch, "Extended target tracking with a lidar sensor using random matrices and a gaussian processes regression model," in *2023 26th International Conference on Information Fusion (FUSION)*, 2023, pp. 1–8.
- [15] K. A. Ruud, E. F. Brekke, and J. Eidsvik, "Lidar extended object tracking of a maritime vessel using an ellipsoidal contour model," in *2018 Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, 2018, pp. 1–6.
- [16] F. Govaers, "On independent axes estimation for extended target tracking, Bonn, Germany," in *Proc. IEEE Sens. Data Fusion: Trends Solut. Appl.*, 2019, pp. 1–6.
- [17] S. Steuernagel, K. Thormann, and M. Baum, "Modeling contour measurements of elliptical extended objects via gaussian spatial distributions," in *2024 Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, 2024, pp. 1–8.
- [18] P. Hoher, J. Reuter, F. Govaers, and W. Koch, "Tracking of partially visible elliptical objects with a lidar sensor using random matrices and a virtual measurement model," in *2022 Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, 2022, pp. 1–6.
- [19] H. Pan, D. Liu, J. Ren, T. Huang, and H. Yang, "Lidar-imu tightly-coupled slam method based on iekf and loop closure detection," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 6986–7001, 2024.
- [20] D. Yanabe, S. Kanoga, V. Frémont, and T. Murakami, "Nonlinear dynamical system identification under external disturbances by maximum a posteriori (map) estimation for robotics," in *2024 IEEE 18th International Conference on Advanced Motion Control (AMC)*, 2024, pp. 1–5.
- [21] X. Zhang, M. N. El Korso, and M. Pesavento, "Maximum likelihood and maximum a posteriori direction-of-arrival estimation in the presence of sirp noise," in *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2016, pp. 3081–3085.
- [22] Y. Yang, X. Liu, X. Liu, Y. Guo, and W. Zhang, "Variational adaptive Im-iekf for full state navigation system of wind disturbance and observability analysis," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–12, 2022.
- [23] F. S. Cattivelli and A. H. Sayed, "Diffusion strategies for distributed Kalman filtering and smoothing," *IEEE Trans. Automat. Contr.*, vol. 55, no. 9, pp. 2069–2084, Sep. 2010.
- [24] A. H. Sayed, "Diffusion adaptation over networks," *Academic Press Library in Signal Processing*, vol. 3, pp. 323–453, 2014.
- [25] F. S. Cattivelli and A. H. Sayed, "Diffusion LMS strategies for distributed estimation," *IEEE Trans. Signal Process.*, vol. 58, no. 3, pp. 1035–1048, Mar. 2010.